# Stochastic Models of the Limit Order Book

## Order flow, queues, and the liquidity provider's problem

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# What this deck covers

1. **The problem.** What a limit order book is, and what a liquidity provider (LP) fights against.
2. **The model toolkit.** Queueing, Hawkes, and propagator models, each stated with its key result.
3. **Order flow.** How imbalance at one level and at several levels moves the price.
4. **The LP's problems now.** Markouts, queue priority, latency, quoting with signals, and liquidity in automated market makers (AMMs).
5. **The frontier, 2023 to 2026.** Generative and foundation models, learning to make markets, rough volatility.
6. **Open problems.**

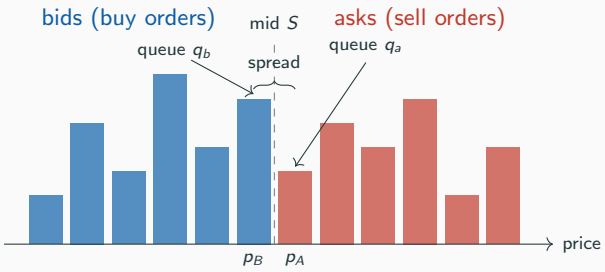
**How to read the numbers**

Every figure and number comes from open code in this repository. Each closed-form result is checked against an independent simulation, and every estimate carries a standard error.

All rate parameters are illustrative. Nothing here is fitted to market data or meant as trading advice.

# The problem

# The limit order book




- Prices sit on a grid with tick size $\delta$ (currency).
- **Limit order**: rests in a first-in, first-out (FIFO) queue at a price.
- **Market order**: trades at once against the best opposite queue.
- **Cancellation**: removes a resting order.
- Best bid $p_B$, best ask $p_A$, mid $S = (p_A + p_B)/2$, spread $s = (p_A - p_B)/\delta$ ticks, best queue sizes $q_b, q_a$ (orders).

The price is an *output* of the queueing system that order flow drives.

# A liquidity provider's P&L, exactly

Let fill $j$ happen at time $t_j$ and price $p_j$, with $\varepsilon_j = +1$ if the LP sold and $-1$ if it bought. Inventory is $y_t = -\sum_{t_j < t} \varepsilon_j$ (shares). Starting flat, the marked-to-market P&L $\Pi_T$ (currency) is

$$\Pi_T \;=\; \underbrace{\sum_j \varepsilon_j \,(p_j - S_{t_j})}_{\text{spread captured } \geq 0} \;+\; \underbrace{\int_0^T y_{t^-}\,\mathrm{d}S_t}_{\text{inventory P\&L}}.$$

This is an identity, not a model. Fill by fill, at a horizon $h$ (s) after the fill:

$$\varepsilon_j\,(p_j - S_{t_j+h}) = \underbrace{\varepsilon_j\,(p_j - S_{t_j})}_{\text{half-spread}} - \underbrace{\varepsilon_j\,(S_{t_j+h} - S_{t_j})}_{\text{adverse selection}}.$$

The left side is the **markout** $M_j(h)$. LPs win the spread on every fill and lose it back when fills predict the next move. Every model in this deck is, for an LP, a model of one of these two terms.

# Five forces on a liquidity provider

| Force | What it does | Where in this deck |
|---|---|---|
| Spread capture | Earns half the spread per passive fill | queue models, markouts |
| Adverse selection | Fills arrive just before the price moves against the LP | markouts, toxic flow, LVR |
| Inventory risk | Holding $y \neq 0$ exposes the LP to $\sigma$ | Avellaneda-Stoikov, GLFT |
| Queue priority | FIFO: the front of the queue fills first and is less toxic | queue position value |
| Latency | Stale quotes are picked off in races | Budish et al., Aquilina et al. |

Fees and rebates shift these trade-offs but do not change their structure. They are left out of the models here.

# Facts the models must reproduce

1. **Order signs have long memory.** The sign autocorrelation decays as a power law, $C_\varepsilon(\ell) \sim \ell^{-\gamma_\varepsilon}$ with $0 < \gamma_\varepsilon < 1$, over thousands of trades (Lillo and Farmer, 2004).
2. **Yet prices are close to unpredictable.** Impact must decay to offset the memory (Bouchaud et al., 2004).
3. **Impact is concave in size.** A metaorder of $Q$ shares moves the price by about $Y\sigma_D\sqrt{Q/V_D}$, with daily volatility $\sigma_D$, daily volume $V_D$ and $Y$ of order one (Tóth et al., 2011; Donier et al., 2015).
4. **Order flow is close to critical.** Fitted Hawkes branching ratios are often near one (Hardiman et al., 2013).
5. **Imbalance predicts the next move.** Order flow imbalance explains a large share of short-horizon price changes (Cont et al., 2014), and deep models find the same price formation mechanism across stocks (Sirignano and Cont, 2019).
6. **Speed is worth money.** Latency races are frequent and a sizeable part of trading costs (Aquilina et al., 2022).

# Notation used throughout

| Symbol | Meaning | Unit | Symbol | Meaning | Unit |
|---|---|---|---|---|---|
| $t$ | time | s | $\lambda$ | limit order rate | $\mathrm{s}^{-1}$ |
| $\delta$ | tick size | currency | $\mu$ | market order rate | $\mathrm{s}^{-1}$ |
| $S$ | mid price | currency | $\theta$ | cancel rate per order | $\mathrm{s}^{-1}$ |
| $s$ | spread | ticks | $\sigma$ | volatility | cur. $\mathrm{s}^{-1/2}$ |
| $q_b, q_a$ | best bid and ask queue | orders | $y$ | LP inventory | shares |
| $i$ | distance to other side | ticks | $M(h)$ | markout at horizon $h$ | ticks |

Order counts are numbers of unit-size orders; $\mu$ is per side; cur. is currency. Prices in ticks are dimensionless. Each slide defines its local symbols.

# The model toolkit

# Model families at a glance

| Family | What drives the rates | Key result | Checked here |
|---|---|---|---|
| Zero intelligence, CST | distance to opposite quote | depth and spread set by $\lambda/\theta$, $\mu/\lambda$ | depth $\to \lambda/\theta$ far out |
| Best queues (CdL) | constants | $p_{\uparrow}(q_b, q_a)$; $1/t$ duration tail; $\sigma^2 \propto \lambda\delta^2$ | MC and exact solve |
| Queue-reactive (HLR) | queue size | stationary law in product form | TV distance 0.0031 |
| Hawkes | past events | $\bar{\Lambda} = (I - G)^{-1}\Lambda^0$; signature plot | MLE, KS, $C(h)$ |
| Propagator | past trade signs | diffusive only at $\zeta_c = (1 - \gamma_\varepsilon)/2$ | slopes $\pm 0.01$ |
| Market making | control of quotes | AS and exact GLFT quotes | CE ranking |

CST: Cont et al. (2010). CdL: Cont and de Larrard (2013). HLR: Huang et al. (2015). GLFT: Guéant et al. (2013). CE: certainty equivalent. MC: Monte Carlo. TV: total variation. MLE: maximum likelihood. KS: Kolmogorov-Smirnov test.

# Zero intelligence and the Markovian book

State: signed queue sizes $X_p$ on a price grid. With $i$ the distance (ticks) to the *opposite* best quote:

- limit orders at distance $i$: rate $\lambda(i)$
- market orders: rate $\mu$ per side
- each resting order cancels at rate $\theta(i)$

Zero intelligence (Smith et al., 2003): $\lambda, \theta$ constant. CST: $\lambda(i) = \lambda_1 i^{-a}$, with $\lambda_1$ ($\mathrm{s}^{-1}$) the rate at distance one and $a$ a dimensionless exponent.

**Dimensional analysis.** With prices in ticks and counts in orders, every stationary statistic depends only on

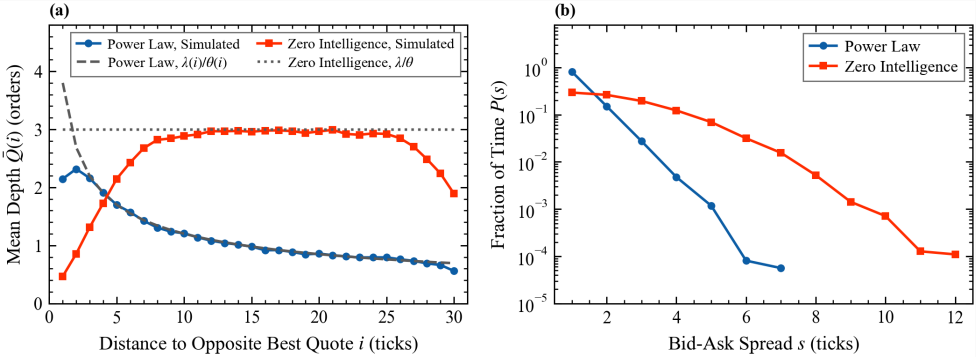
$$\lambda/\theta \text{ (depth)}, \quad \mu/\lambda \text{ (price scale)}, \quad L \text{ (window)}.$$

Why it matters: Farmer et al. (2005) found that this model, with rates measured per stock, explains most of the cross-sectional variation in mean spread on the London Stock Exchange.

What it misses: rates depend on the state and on history; order signs have no memory; no information.

Simulated exactly by the Gillespie algorithm on a moving grid.

# Markovian book: depth profile and spread




- Far from the quotes, depth $\bar{Q}(i)$ (orders) follows the M/M/$\infty$ value $\lambda(i)/\theta(i)$; near the quotes, market orders thin the book.
- Mean spread: $1.226 \pm 0.006$ ticks (power law) against $2.610 \pm 0.028$ ticks (zero intelligence, $\mu/\lambda$ six times larger). Standard errors from 20 batch means over 300,000 events.

# Best queues: direction and timing of the next price move

Two best queues with one-tick spread; each grows at rate $\lambda$ and shrinks at rate $\nu$ (market orders plus cancellations, $\mathrm{s}^{-1}$). The price moves up when the ask queue empties first (Cont and de Larrard, 2013).

First passage of a queue of size $q$ to zero, density ($\mathrm{s}^{-1}$):

$$f_q(t) = \frac{q}{t}\left(\frac{\nu}{\lambda}\right)^{q/2} I_q\big(2\sqrt{\lambda\nu}\,t\big)\, e^{-(\lambda+\nu)t},$$

with $I_q$ the modified Bessel function. Then

$$p_\uparrow(q_b, q_a) = \int_0^\infty f_{q_a}(t)\, P(\tau_{q_b} > t)\, \mathrm{d}t.$$

Balanced case $\lambda = \nu$, large queues:

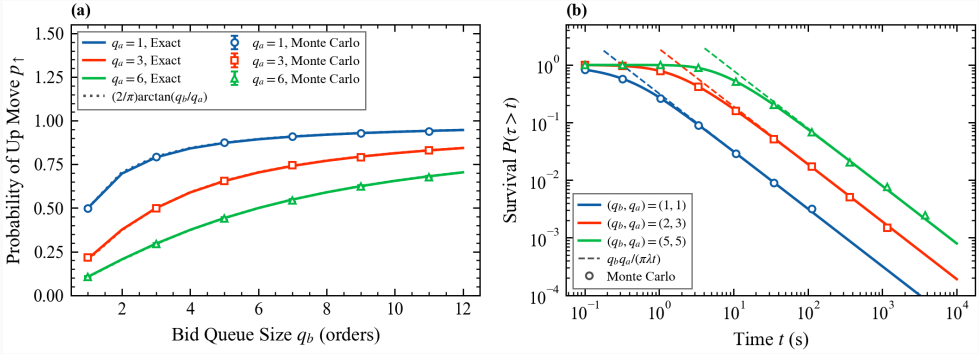
$$p_\uparrow \approx \tfrac{2}{\pi} \arctan(q_b/q_a).$$

Time to the next move:
$P(\tau > t) \sim q_b q_a/(\pi\lambda t)$, so the **mean duration is infinite**.

Price diffusion, with $M_F$ the mean product of regenerated queue sizes (orders$^2$):

$$\sigma^2 = \pi\lambda\delta^2/M_F \quad \text{(time scale } t\, n\log n\text{)}.$$

# Best queues: closed forms against simulation

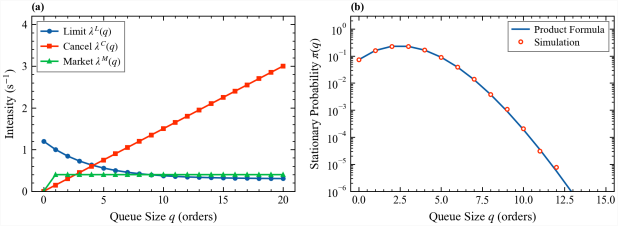



- Monte Carlo (20,000 paths per point) matches the closed form within 2.4 standard errors; an exact sparse linear solve agrees to $2 \times 10^{-3}$.
- The arctangent law is off by 0.0017 at $(2, 5)$ and $2.5 \times 10^{-5}$ at $(40, 20)$. Imbalance alone is a strong predictor, which is why micro-prices work (Stoikov, 2018).

# Queue-reactive models

Huang et al. (2015) estimate every rate as a function of the queue size $q$ (in average event sizes): $q \to q+1$ at $\lambda^L(q)$, $q \to q-1$ at $\lambda^C(q) + \lambda^M(q)$. Model I (independent queues) has the product-form stationary law

$$\pi(q) = \pi(0) \prod_{j=1}^{q} \lambda^L(j-1) / \left[\lambda^C(j) + \lambda^M(j)\right].$$




Illustrative rates (a); product formula against one simulated queue of 314,592 events (b): total variation distance 0.0031. Models II and III couple the best queues and move a reference price; Wu et al. (2019) add Hawkes terms.

# Hawkes processes: order flow excites itself

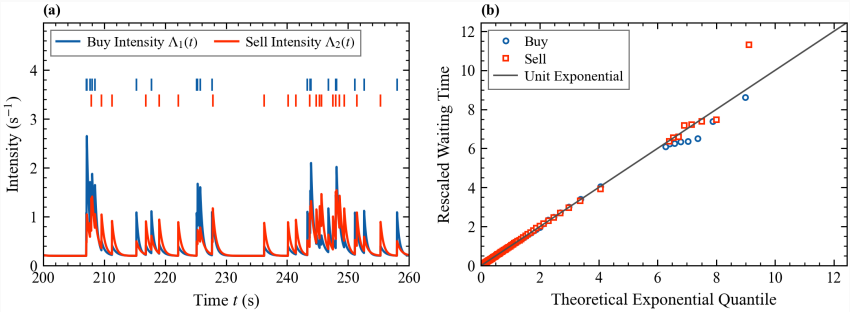
For event types $u, v$ (buy, sell, cancel, ...) the intensity ($\mathrm{s}^{-1}$) is

$$\Lambda_u(t) = \Lambda_u^0 + \sum_v \sum_{t_j^v < t} \alpha_{uv}\, e^{-\beta_{uv}(t - t_j^v)},$$

with baseline $\Lambda_u^0$, jump $\alpha_{uv}$ and decay $\beta_{uv}$, all in $\mathrm{s}^{-1}$. The branching matrix $G_{uv} = \alpha_{uv}/\beta_{uv}$ counts direct offspring; stationarity needs spectral radius $\rho(G) < 1$, and then $\bar{\Lambda} = (I - G)^{-1}\Lambda^0$.

- Fitted order flow is often **near critical**, $\rho(G) \approx 1$ (Hardiman et al., 2013), and the kernels decay slowly, so they must be estimated from about 10 $\mu$s to 100 s (Bacry et al., 2016).
- Exact likelihood in one pass for exponential kernels; residuals $\mathcal{A}_u(t_j) - \mathcal{A}_u(t_{j-1})$ of the compensator $\mathcal{A}_u$ are i.i.d. unit exponential under the true model.
- Nearly unstable Hawkes processes have diffusive scaling limits (Jaisson and Rosenbaum, 2015); with heavy-tailed kernels the limits are **rough**, which gives rough volatility with leverage from microstructure (El Euch et al., 2018; Wang et al., 2026).
- Estimation pitfalls (seasonality, regime shifts, one-scale kernels): deep dive 4.

# Hawkes: fitting buy and sell market orders




Bivariate exponential Hawkes model of buys and sells, $T_{\text{obs}} = 10{,}000$ s of data: 3,972 buys and 4,523 sells, true spectral radius $\rho(G) = 0.518$. (a) Intensities over one minute; ticks mark events.

- MLE recovers all ten parameters within 2.23 standard errors (observed information).
- (b) Rescaled waiting times are unit exponential: Kolmogorov-Smirnov $p = 0.94$ and 0.99.

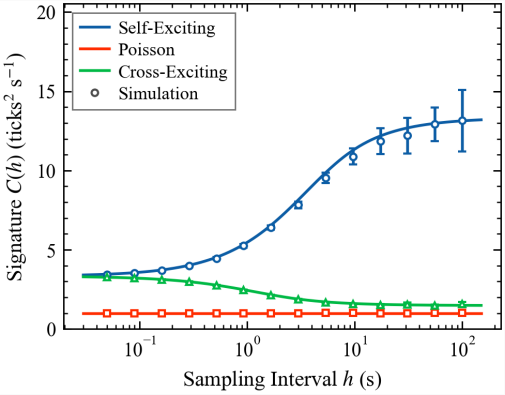
# Hawkes price model: microstructure noise

Up ticks $N_+$ and down ticks $N_-$ form a symmetric Hawkes process: self jump $\alpha_s$, cross jump $\alpha_c$, decay $\beta$ (Bacry et al., 2013). With $g = (\alpha_s - \alpha_c)/\beta$ and $\eta = \beta(1-g)$ ($s^{-1}$), the variance per unit time at sampling interval $h$ (s) is

$$C(h) = 2\bar{\Lambda}\left[\frac{1}{(1-g)^2} + \left(1 - \frac{1}{(1-g)^2}\right)\frac{1-e^{-\eta h}}{\eta h}\right]$$

in ticks$^2$ $s^{-1}$. Cross-excitation ($g < 0$) gives the falling signature plot of tick data: bid-ask bounce. Derivation in the backup slides.

Simulation matches within 3.1 standard errors over 42 points (10 replicates each).



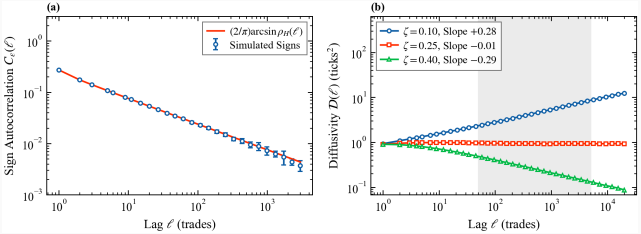
# The propagator model: memory and efficiency

Trade time $n$, sign $\varepsilon_n = \pm 1$, price in ticks (Bouchaud et al., 2004):

$$P_n = \sum_{m<n} \Gamma_0 \, (n-m)^{-\zeta} \, \varepsilon_m,$$

with impact amplitude $\Gamma_0$ (ticks) and decay exponent $\zeta$.

If $C_\varepsilon(\ell) \sim \ell^{-\gamma_\varepsilon}$, then $\mathcal{D}(\ell) = \mathrm{Var}(P_{n+\ell} - P_n)/\ell \sim \ell^{1-\gamma_\varepsilon-2\zeta}$, so prices diffuse only at $\zeta_c = (1-\gamma_\varepsilon)/2$: impact decay offsets sign memory.




Signs: sign of fractional Gaussian noise, $\gamma_\varepsilon = 0.50$, $2^{20}$ trades, 8 replicates. Fitted slopes $0.284 \pm 0.010$, $-0.007 \pm 0.008$, $-0.286 \pm 0.006$ against theory 0.30, 0.00, $-0.30$.

# Large orders: the square-root law and the latent book

A metaorder of $Q$ shares, executed over hours, moves the price by

$$I(Q) \approx Y \, \sigma_D \sqrt{Q/V_D},$$

with daily volatility $\sigma_D$ (fraction of price), daily volume $V_D$ (shares) and a dimensionless $Y$ of order one (Tóth et al., 2011).

- The visible book cannot explain it: it holds a small fraction of daily volume. Donier et al. (2015) model a **latent** order book whose density is linear near the price. This gives the square root, contains the propagator model, and allows no price manipulation.
- For an LP the message is that impact has two parts: a transient mechanical part that reverts, and a permanent informational part. Markouts at long horizons measure the second.

The square-root law is an empirical result; this repository does not test it.

# Order flow

# Order flow imbalance: linear impact at short horizons

From best-quote snapshots $n-1 \to n$ (Cont et al., 2014):

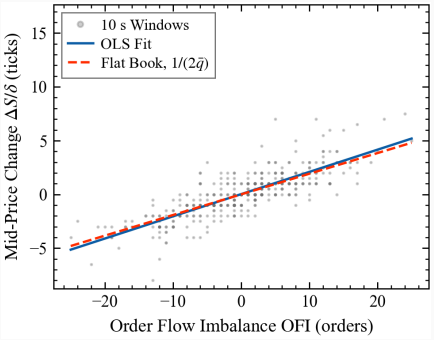
$$e_n = \mathbf{1}_{\{p_{B,n} \geq p_{B,n-1}\}} q_{b,n} - \mathbf{1}_{\{p_{B,n} \leq p_{B,n-1}\}} q_{b,n-1} - \mathbf{1}_{\{p_{A,n} \leq p_{A,n-1}\}} q_{a,n} + \mathbf{1}_{\{p_{A,n} \geq p_{A,n-1}\}} q_{a,n-1}.$$

New bids, cancelled asks and market buys push $e_n$ up. Summed over a window, $\mathrm{OFI}$ (orders) explains the mid change linearly:

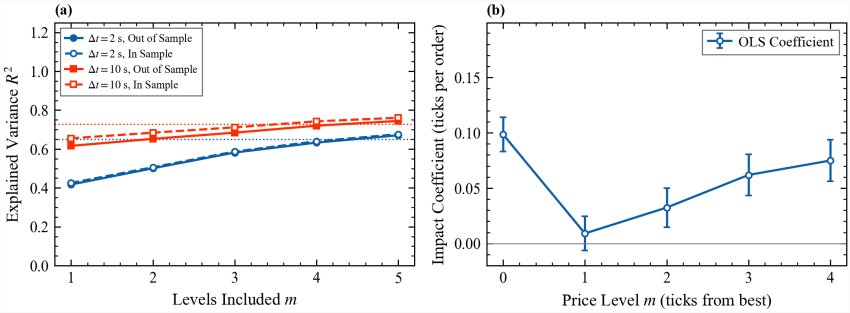
$$\Delta S/\delta = \kappa\, \mathrm{OFI} + \xi, \qquad \kappa \approx 1/(2\bar{q}),$$

with $\bar{q}$ the mean best-queue depth: a flat book of depth $\bar{q}$ moves one quote $e/\bar{q}$ ticks and the mid half as much.

Simulated book: $\kappa = 0.2067 \pm 0.0072$ against $1/(2\bar{q}) = 0.1921$; $R^2 = 0.66$ over 412 windows of 10 s.



# Beyond the best quotes: multi-level and integrated OFI




**Multi-level OFI**: the same formula at each level $m$ ticks from the best (Xu et al., 2019). **Integrated OFI**: levels combined by the first principal component of depth-normalized OFIs (Cont et al., 2023); dotted lines in (a).

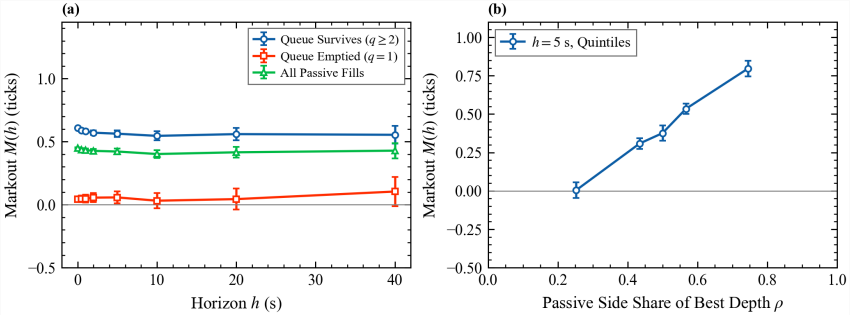
Out-of-sample $R^2$, $\Delta t = 10$ s: best level 0.62, 5 levels 0.74, integrated 0.73. At $\Delta t = 2$ s: 0.42, 0.67, 0.65. Deeper queues decide how far the price moves once the best queue empties.

# What the data say about order flow

- **Integrated OFI beats best-level OFI** for contemporaneous impact. Once multi-level OFI is used, **contemporaneous cross-impact** between stocks adds no explanatory power, but lagged cross-asset OFI does forecast returns at short horizons (Cont et al., 2023).
- **Imbalance is a signal with a finite life.** It predicts future price moves and mean-reverts (Lehalle and Neuman, 2019). A signal's half-life sets how an LP should use it (next section).
- **Deep models of the raw book**: convolutional and recurrent networks on the book forecast short-term moves out of sample (Zhang et al., 2019); pooled training across stocks beats per-stock models, which points to a **universal** price formation mechanism (Sirignano and Cont, 2019); deep OFI features forecast a term structure of returns at several horizons (Kolm et al., 2023).
- **Hawkes with state dependence**: market order and mid-change rates depend mainly on volume imbalance, while limit and cancel rates do not (Wu et al., 2019).

# The liquidity provider's problems now

# Markouts: which fills are toxic




Markout of a passive fill at horizon $h$ (s):
$M(h) = \varepsilon\,(p - S_{t+h})$ ticks, with $\varepsilon = +1$ for a passive sale. Markovian book, 16,503 fills in 600,000 events; bars at two standard errors.

- (a) 0.28 of fills **empty the queue**; they keep $0.04 \pm 0.01$ ticks against $0.61 \pm 0.00$.
- (b) Thin passive side ($\rho$ low): $0.00 \pm 0.03$; thick side: $0.80 \pm 0.03$ ticks.
- No informed traders here: queue mechanics alone create adverse selection.

# Queue position: a model with informed sweeps

We hold one sell order at the best ask with $a$ orders ahead and $b$ behind; the bid queue holds $c$. Rates ($s^{-1}$):

- market buys $\mu$ hit the *front* of the queue;
- limit sells $\lambda$ join *behind* us;
- each other order cancels at $\theta$;
- informed **sweeps** at rate $\nu$ take the whole queue and move fair value $J$ ticks; a sweep of the bid moves the price away.

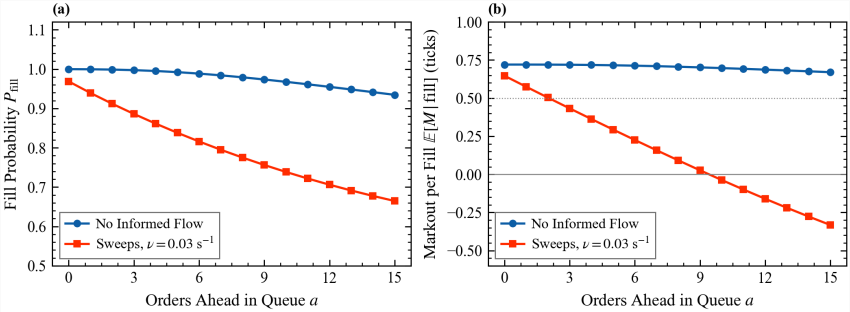
Markout per fill, against the mid after the next price change:

$$\tfrac{1}{2}-\mathrm{AS}, \quad \mathrm{AS} = \begin{cases} 1 & \text{queue emptied} \\ 2P_{\uparrow}(c,b) - 1 & b > 0 \\ J & \text{sweep.} \end{cases}$$

Here $P_{\uparrow}(c, b)$ is the probability that the ask queue empties before the bid queue. Fill probability and value solve a sparse linear system on $(a, b, c)$; checked by Monte Carlo and truncation.

Idea from Moallemi and Yuan (2017): small orders fill the front, sweeps fill everyone.

# The value of queue position




Queue of 16 orders including ours, bid queue 10; $\lambda = 1$, $\mu = 0.9$, $\theta = 0.01$ s$^{-1}$; sweeps $\nu = 0.03$ s$^{-1}$ with $J = 2$ ticks.

- Front: fill probability 0.97, 0.65 ticks per fill. Back: 0.66 and $-0.33$.
- Joining behind $a = 10$ loses money. Without sweeps the curve is flat (0.72 to 0.67).

# Latency: the race for stale quotes

When public information arrives, fast traders race to trade against quotes that have not yet moved, and LPs race to cancel them.

**Measured** on London Stock Exchange message data, including failed orders (Aquilina et al., 2022):

- about one race per minute per FTSE 100 symbol, lasting microseconds;
- about 20% of trading volume;
- about a third of price impact and of effective spread;
- eliminating races would cut the cost of liquidity by about 17%.

**Design response.** Continuous trading makes speed a prisoner's dilemma; frequent batch auctions turn competition on speed into competition on price (Budish et al., 2015).

**In the models.** Races play the role of the sweeps on the previous slide: a stale quote is filled exactly when it should not be. Speed buys a lower effective $\nu$.

# Optimal quotes with inventory limits: the exact solution

Mid $\mathrm{d}S = m\,\mathrm{d}t + \sigma\,\mathrm{d}W$; a quote $d$ away fills at rate $Ae^{-kd}$; the LP maximizes $\mathbb{E}[-e^{-\gamma(x_T + y_T S_T)}]$ with $|y| \le Q$ (shares). Here $m$ is a drift (ticks $\mathrm{s}^{-1}$), $A$ ($\mathrm{s}^{-1}$) and $k$ ($\mathrm{tick}^{-1}$) set the fill rate, $\gamma$ ($\mathrm{tick}^{-1}$) is risk aversion.

**Avellaneda-Stoikov** expand in inventory (Avellaneda and Stoikov, 2008):

$$r_t = S_t - y\gamma\sigma^2(T-t),$$

$$d^a + d^b = \gamma\sigma^2(T-t) + \tfrac{2}{\gamma}\ln\left(1+\tfrac{\gamma}{k}\right).$$

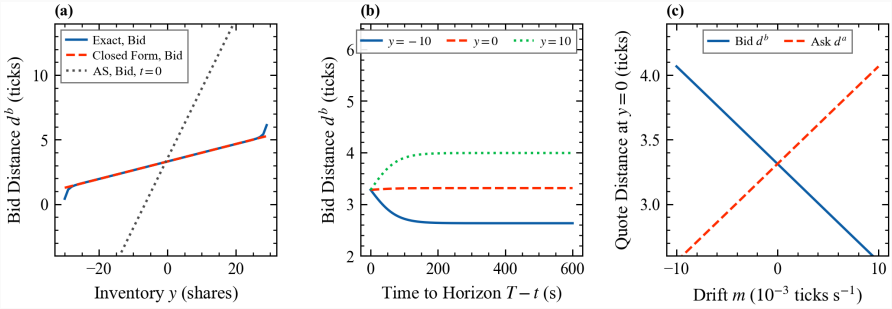
Accurate only near the horizon.

**GLFT** make the Hamilton-Jacobi-Bellman (HJB) equation linear (Guéant et al., 2013): $v(t) = e^{-M(T-t)}\mathbf{1}$ with tridiagonal $M$,

$$d^b(t,y) = \tfrac{1}{k}\ln\tfrac{v_y}{v_{y+1}} + \tfrac{1}{\gamma}\ln\left(1+\tfrac{\gamma}{k}\right),$$

and the mirror formula for $d^a$. A drift enters $M$ only through its diagonal. Details in the backup slides.

# Exact quotes against approximations

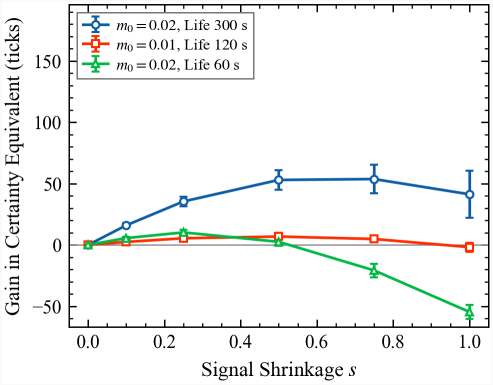



(a) Long-horizon exact quotes match the closed form within 0.001 ticks near $y = 0$; the AS expansion at $t = 0$ overreacts. (b) Quotes are stationary after about 200 s. (c) Drift tightens one side and widens the other.

| Strategy | Mean $\Pi_T$ | Std $\Pi_T$ | CE |
|---|---|---|---|
| Exact GLFT | $1319.2 \pm 1.6$ | 73.6 | 1291.9 |
| AS approximation | $1301.0 \pm 1.5$ | 68.5 | 1277.7 |
| Symmetric | $1324.1 \pm 2.7$ | 120.4 | 1227.7 |

2,000 paths; CE: certainty equivalent (ticks). GLFT example parameters: $\sigma = 0.3$ tick s$^{-1/2}$, $A = 0.9$ s$^{-1}$, $k = 0.3$ tick$^{-1}$, $\gamma = 0.01$ tick$^{-1}$, $T = 600$ s, $Q = 30$.

# Quoting with a short-term signal




Drift $\pm m_0$ (ticks $\mathrm{s}^{-1}$) flips sign at random, mean life $1/\kappa$ (s). Exact GLFT treats a drift as *permanent*, so the LP plugs in $s\,m(t)$ with shrinkage $s \in [0, 1]$.

- Life 300 s: best $s = 0.75$, gain $53.7 \pm 5.8$ ticks of CE.
- Life 60 s: raw use **loses** $-54.6 \pm 2.9$; $s = 0.25$ gains $10.2 \pm 1.0$.

The signal's life relative to inventory turnover sets how much to use. Paired runs, bootstrap s.e.

# Toxic flow, brokers and internalization

A dealer or broker sees *who* trades. That changes the problem from "how wide" to "with whom".

- Cartea and Sánchez-Betancourt (2025): a broker quotes to informed and uninformed clients and trades in the lit market. It learns a trend signal from the informed flow and decides what to internalize and what to pass on.
- Cartea et al. (2023): predicting which client trades are toxic, with neural networks trained by a fast online Bayesian method; on foreign exchange data this beats logistic regression and random forests in profit.
- Markouts are the label; the forecast must come before the fill. The queue-position and markout results above show that the label depends on queue state as well as on the client.

# Liquidity in an AMM: loss-versus-rebalancing

An AMM LP holds $x(P)$ units of the risky asset and is arbitraged to the exchange price $P$, a geometric Brownian motion with volatility $\sigma$ (day$^{-1/2}$). Against a strategy that holds the same $x(P)$ but trades at $P$, it loses at the rate (Milionis et al., 2022)

$$\frac{\mathrm{d}\,\mathrm{LVR}}{\mathrm{d}t} = \frac{\sigma^2 P^2}{2}\,|x'(P)| = \frac{\sigma^2}{8}\,V \quad \text{(constant product, value } V = 2L\sqrt{P}\text{)}.$$

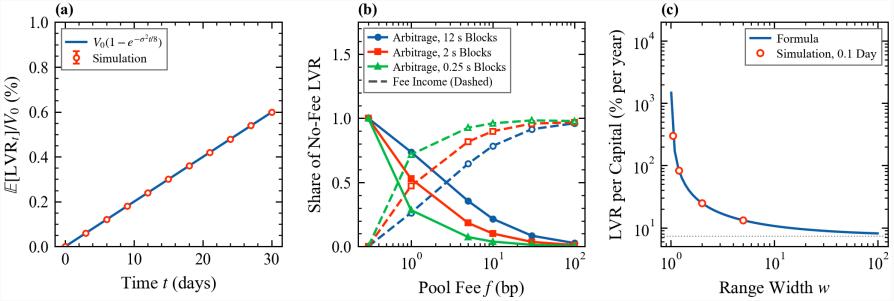
- Exact mean: $\mathbb{E}[\mathrm{LVR}_T] = V_0(1 - e^{-\sigma^2 T/8})$.
- Fees $f$ and discrete blocks create a no-trade band; arbitrage profit falls and part of LVR returns as fees (Milionis et al., 2023).

- Concentrated liquidity on $[P/w, Pw]$ (derived here, backup slide):

$$\frac{\mathrm{LVR}}{V} = \frac{\sigma^2}{8\,(1 - w^{-1/2})}.$$

LP losses here are adverse selection, the same economics as stale quotes in a book.

# LVR: fees, block time and concentration




(a) $\sigma = 0.04$ day$^{-1/2}$, about 76% a year: LVR costs 7.3% of capital per year; 30-day simulation $0.598 \pm 0.001\%$ against exact 0.598%.

(b) At 5 bp and 12 s blocks, arbitrage keeps 0.35 and fees return 0.65 of no-fee LVR; faster blocks cut arbitrage further. Zero fee drawn at 0.3 bp. (c) A $w = 1.1$ range pays $21.5\times$ the full-range LVR per unit capital.

# Measuring it on real data: pitfalls

**Markouts**

- Choose the benchmark: mid or micro-price at fill time changes the split between spread and adverse selection.
- Overlapping horizons make fills correlated; use block bootstrap or autocorrelation-robust errors.
- Clock time and event time give different curves; report both.

**Fill probabilities**

- Cancelled orders are censored; use survival analysis. First-passage approximations are poor stand-ins for real executions (Lo et al., 2002).

**Queue position**

- Needs order-level (market-by-order) messages and FIFO bookkeeping; hidden and iceberg orders break it.

**Hawkes and order flow**

- Timestamp resolution and same-time bursts bias kernel estimates at short lags.
- Remove intraday seasonality before fitting; otherwise it looks like excitation.

**Everything**

- Fit out of sample and by regime; report uncertainty with every number.

# What a liquidity provider should take away

1. **Measure markouts, not fill rates.** Spread capture is visible on every fill; adverse selection shows up only in $M(h)$. Condition it on queue state and imbalance.
2. **Priority is an asset with a price.** The front of a queue is worth more than the back, and the gap grows with the rate of informed sweeps. Joining a long queue late can have negative value.
3. **Use the exact control solution.** With inventory limits, the GLFT matrix exponential is cheap and beats the AS expansion; the stationary quotes are a single eigenvector.
4. **Shrink signals by their life.** A permanent-drift model fed a short-lived signal loses money; the right shrinkage turns the same signal into a gain.
5. **Same economics on chain.** AMM LPs pay LVR to arbitrageurs as book LPs pay latency arbitrageurs; fees, block time and range width set the bill.

# The frontier, 2023 to 2026

# Generative models of the order book

The shift: from models of a few summary statistics to models of the **full message stream**.

| Work | What it does |
|---|---|
| Nagy et al. (2023) | Token-level autoregressive model of LOBSTER messages with a deep state-space network; generated order flow drives a book simulator. |
| LOB-Bench (Nagy et al., 2025) | Benchmark for generated LOB data: spreads, volumes, imbalance, inter-arrival times, a discriminator score, and **market impact** metrics. Autoregressive models beat a (C)GAN and a parametric model. |
| LOBERT (Linna et al., 2025) | Encoder-only foundation model; one token per full message, with continuous price, volume and time. |
| TradeFM (Kawawa-Beaudan et al., 2026) | 524M-parameter transformer on trade events from over 9,000 stocks; one model across assets. |
| Mohl et al. (2026) | Probes a Mamba state-space model: OFI is linearly decodable from its states ($R^2 \approx 0.59$), transfers to unseen stocks, and steering it changes predictions. |

Open question for LPs: do these simulators respond *causally* to a new order the way a real book does? Matching marginal statistics is not enough to backtest a quoting strategy.

# Learning to make markets

**Reinforcement learning.** Market making framed as a Markov decision process; RL agents tend to beat analytical benchmarks on risk-adjusted return in simulation (Gašperov et al., 2021). The simulator is the weak link: an agent learns the flaws of its environment.

**Online learning with order book feedback.** Trader valuations are seen only when no trade occurs.

- Regret improves from $\tilde{O}(T^{2/3})$ to $\tilde{O}(\sqrt{T})$ with high probability; sublinear regret is impossible if prices and valuations are both adversarial (Vischi et al., 2026).
- The value of observing the book is made precise by an action-dependent feedback model (Maran and Restelli, 2026).

**Signals inside the control problem.** Optimal trading with a Markov signal and transient impact has explicit solutions (Lehalle and Neuman, 2019). Exact market making with a drift is a linear ODE system (Guéant et al., 2013); next slides use it.

**Brokers and toxic flow.** A broker can learn a trend signal from informed clients' flow and decide what to internalize (Cartea and Sánchez-Betancourt, 2025); online Bayesian neural networks predict which client trades are toxic (Cartea et al., 2023).

# Microstructure as the source of volatility

- Hawkes order flow near criticality ($\rho(G) \to 1$) has volatility-like scaling limits (Jaisson and Rosenbaum, 2015); with heavy-tailed kernels these limits are **rough** (Hurst exponent below one half).
- No-arbitrage, buy-sell asymmetry and metaorder splitting give a rough Heston model with leverage (El Euch et al., 2018); recent work adds common price and volatility jumps (Wang et al., 2026).
- For an LP this links two time scales: the branching ratio estimated from milliseconds of order flow controls the volatility it must hedge over minutes and hours.

**Market design**

Continuous books create speed races; frequent batch auctions turn competition on speed into competition on price (Budish et al., 2015). AMMs remove the queue entirely but leave LPs exposed to arbitrage (loss-versus-rebalancing).

# Open problems

# Open problems: models

*This list is our synthesis of the gaps visible in the work above, not a list taken from one source.*

1. **One model for three facts.** Queue-level dynamics, long-memory order signs, and square-root metaorder impact are each explained by a different model family. No tractable model yet produces all three from one mechanism.
2. **State plus history.** Queue-reactive models use the state, Hawkes models the past. Their union exists (Wu et al., 2019), but estimation with power-law kernels, many event types and order sizes is still hard.
3. **Nonstationarity.** All models here are stationary. Intraday seasonality, news, and regime shifts change rates by large factors. Detecting regimes early, and quoting through them, is open.
4. **Causal validation of simulators.** Generative models match statistics; an LP needs the *response* to its own orders. Benchmarks now include impact metrics (Nagy et al., 2025), but no standard test checks counterfactual fills and markouts.
5. **Hidden and latent liquidity.** The visible book is a small slice of supply and demand (Donier et al., 2015). Inferring the latent book online, from the visible book and fills, is open.

# Open problems: liquidity provision

6. **Queue value in equilibrium.** Queue position has value (previous section). But every LP competes for the front; the equilibrium value of priority, and of speed spent to obtain it, needs game-theoretic models tied to measured queue dynamics.
7. **Signals with finite life in optimal control.** Exact market making assumes a permanent drift. Short-lived signals need a larger state (signal plus inventory). We show that naive use hurts and that shrinkage helps; the optimal map from signal half-life to quotes is open in closed form.
8. **Real-time toxicity with uncertainty.** Markouts reveal toxicity only after the fact. Predicting it per fill, with calibrated uncertainty, under feedback (toxic clients adapt), is open (Cartea et al., 2023).
9. **Multi-asset and cross-venue.** Lagged cross-impact forecasts returns (Cont et al., 2023), and multi-asset quotes beat independent ones (Guéant, 2017). Fragmented venues, latency between them, and inventory netting across assets remain hard.
10. **LOB versus AMM design.** Fees and block times trade LP losses against trader costs (Milionis et al., 2023). Which design minimizes total cost of liquidity for a given asset is an open design question.

# The repository

# Deep dives: one deck per topic

Each deck goes from the question to the model, its statistics, figures from this code, published evidence, and a discussion for liquidity providers. PDFs in `slides/decks/`.

| | Deck | New in the deck |
|---|---|---|
| 1 | The Markovian order book | rate estimation from LOBSTER files; the depth-truncation bias in cancellation rates |
| 2 | The best queues | constant-rate formulas fitted to a state-dependent book; clustered errors for conditional frequencies |
| 3 | Queue-reactive models | nonparametric intensity estimation; a reduced Model III: mean reversion and news |
| 4 | Self-exciting order flow | three ways an exponential fit misreads the branching ratio |
| 5 | Price impact | propagator estimation and truncation bias; latent book: exact linear-to-square-root crossover |
| 6 | Order flow imbalance and the micro-price | micro-price estimation and out-of-sample tests |
| 7 | Optimal market making | fill-curve estimation with censoring; cost of a wrong fill curve |
| 8 | Adverse selection and queue priority | Glosten-Milgrom simulated; markouts; queue value |
| 9 | Liquidity in AMMs | LVR derivation, concentrated liquidity, fees and blocks |

# Everything here is reproducible

**Package** `stochastic_lob`, NumPy and SciPy only:

- `markov_lob`, `best_queues`, `queue_reactive`
- `hawkes`, `impact`, `latent_book`, `microprice`
- `market_making`, `adverse_selection`, `queue_position`, `markouts`, `amm`
- `lobster`: read and write LOBSTER order book files

**Tests** (46): each closed form or estimator against an independent computation (Monte Carlo, exact linear solve, PDE, or a known limit).
**Experiments** (23): one script per figure; numbers go to `results/` as LaTeX macros, and the slides read them.

```
pip install -e .
python -m pytest tests
python experiments/run_all.py
python build_slides.py
```

Fixed random seeds; about 15 minutes on a laptop for all experiments.

# Verification summary

| Result | Check | Agreement |
|---|---|---|
| CdL $p_\uparrow$ closed form | Monte Carlo; sparse solve | $\leq 2.4$ s.e.; $2 \times 10^{-3}$ |
| CdL duration tail $q_b q_a/(\pi\lambda t)$ | exact survival at $10^4$ s | within 0.88% |
| Queue-reactive stationary law | long simulation | TV 0.0031 |
| Hawkes MLE, 10 parameters | known truth | $\lvert z\rvert \leq 2.23$ |
| Hawkes signature plot | 10 replicates, 42 points | $\leq 3.1$ s.e. |
| Propagator, critical $\zeta_c$ | fitted log-log slope | $-0.007 \pm 0.008$ vs 0 |
| Queue value (absorbing chain) | Monte Carlo; truncation | within 4 s.e.; $10^{-3}$ |
| Exact GLFT | ODE residual; closed form at small $y$ | $< 10^{-6}$; 0.001 ticks |
| CPMM LVR $V_0(1 - e^{-\sigma^2 T/8})$ | Monte Carlo | within 4 s.e. |
| Concentrated LVR $\sigma^2/(8(1 - w^{-1/2}))$ | Monte Carlo | within 2% |
| Latent book: linear-to-square-root $F(u)$ | finite-difference PDE | within 0.8% |
| Book rate MLE $\hat{\lambda}(i)$, $\hat{\theta}(i)$ | known truth | $\lvert z\rvert \leq 2.5$ |
| State-dependent $p_\uparrow$ | simulated book, clustered s.e. | $\lvert z\rvert \leq 2.1$ |

s.e.: standard error. TV: total variation distance. $z$: (estimate − truth)/s.e.

# Backup: derivations and model details

# Backup: the P&L identity

Cash after fills: $x_T = \sum_j \varepsilon_j p_j$ (sell: receive $p_j$). Inventory: $y_T = -\sum_j \varepsilon_j$. Then

$$\Pi_T = x_T + y_T S_T = \sum_j \varepsilon_j (p_j - S_{t_j}) - \sum_j \varepsilon_j (S_T - S_{t_j}).$$

Since $y_{t^-} = -\sum_{t_j < t} \varepsilon_j$, the last sum equals $\int_0^T y_{t^-}\, \mathrm{d}S_t$. No model enters. Taking expectations, the inventory term is negative exactly when fills are followed by moves against the new position, which is adverse selection; its variance is inventory risk.

# Backup: arctangent law and duration tail

**Arctangent.** For $\lambda = \nu$ each queue is a symmetric walk with variance rate $2\lambda$ (orders$^2$ s$^{-1}$). For large queues $(q_b, q_a)$ is a planar Brownian motion with independent, equal-variance coordinates, stopped on the axes. The probability $h$ of exiting through $\{q_a = 0\}$ is harmonic with $h = 1$ there and $h = 0$ on $\{q_b = 0\}$. In polar angle $\chi = \arctan(q_a/q_b)$, $h = 1 - 2\chi/\pi$ works, so $p_\uparrow \approx (2/\pi)\arctan(q_b/q_a)$. It fails if $\lambda \neq \nu$ (drift).

**Tail.** By the reflection principle, $P(\tau_q > t) \approx P(|Z| < q/\sqrt{2\lambda t}) \approx q/\sqrt{\pi\lambda t}$ for a standard normal $Z$. Independence gives $P(\tau > t) \approx q_b q_a/(\pi\lambda t)$. With a $1/t$ tail the truncated mean grows like $\log t$, the number of price changes by time $t$ grows like $t/\log t$, and the $n \log n$ time scale of the diffusion limit follows. The renewal step is heuristic; Cont and de Larrard (2013) prove it.

# Backup: signature plot of the Hawkes price

For a stationary linear Hawkes process the spectral density matrix of $\mathrm{d}N$ at angular frequency $\omega_f$ is $\frac{1}{2\pi}[I-\hat{\Phi}]^{-1}\mathrm{diag}(\bar{\Lambda})[I-\hat{\Phi}]^{-*}$ (Hawkes, 1971). The price $\mathbf{w}^\top N$ with $\mathbf{w}=(1,-1)^\top$ uses the eigenvalue $\hat{\psi}=(\alpha_s-\alpha_c)/(\beta+\mathrm{i}\,\omega_f)$, so the increments have density

$$\frac{2\bar{\Lambda}}{2\pi}\Big[1+\frac{\beta^2-\eta^2}{\eta^2+\omega_f^2}\Big], \qquad \eta=\beta-\alpha_s+\alpha_c.$$

In time: covariance density $2\bar{\Lambda}[\delta_D(t)+\frac{\beta^2-\eta^2}{2\eta}e^{-\eta|t|}]$, with $\delta_D$ the Dirac delta. Integrating over $[0,h]^2$ and dividing by $h$ gives $C(h)$. Checks: $C(0^+)=2\bar{\Lambda}$; no excitation gives the Poisson value $2\Lambda^0$; simulation agrees (slide on signature plot).

# Backup: exact market making with inventory limits

Mid $\mathrm{d}S = m\,\mathrm{d}t + \sigma\,\mathrm{d}W$, fills at rate $Ae^{-kd}$ for a quote at distance $d$, utility $-\exp(-\gamma(x_T + y_T S_T))$, inventory $|y| \le Q$ (Guéant et al., 2013). The ansatz

$$u(t,x,y,S) = -\exp\big(-\gamma(x+yS)\big)\, v_y(t)^{-\gamma/k}$$

turns the HJB equation into the linear system $\dot{v}(t) = Mv(t)$, $v(T) = \mathbf{1}$, with $M$ tridiagonal:

$$M_{yy} = \tfrac{k}{2}\gamma\sigma^2 y^2 - km\,y, \qquad M_{y,y\pm1} = -A\big(1+\tfrac{\gamma}{k}\big)^{-(1+k/\gamma)}.$$

So $v(t) = e^{-M(T-t)}\mathbf{1}$ and

$$d^b(t,y) = \tfrac{1}{k}\ln\tfrac{v_y(t)}{v_{y+1}(t)} + \tfrac{1}{\gamma}\ln\big(1+\tfrac{\gamma}{k}\big), \qquad d^a(t,y) = \tfrac{1}{k}\ln\tfrac{v_y(t)}{v_{y-1}(t)} + \tfrac{1}{\gamma}\ln\big(1+\tfrac{\gamma}{k}\big).$$

Long horizons: $v$ aligns with the ground state of $M$. Implemented by diagonalizing $M$ (it is symmetric). Checks: terminal quotes, ODE residual below $10^{-6}$, small-$y$ agreement with the closed form.

# Backup: LVR of concentrated liquidity

For holdings $x(P)$ of the risky asset, the LVR rate is $\frac{\sigma^2 P^2}{2}|x'(P)|$ (Milionis et al., 2022), with $\sigma$ the volatility per $\sqrt{\text{day}}$.

- CPMM, $x = L/\sqrt{P}$: rate $\frac{\sigma^2}{4} L\sqrt{P} = \frac{\sigma^2}{8} V$, with value $V = 2L\sqrt{P}$. Since $\mathbb{E}\sqrt{P_T} = \sqrt{P_0} e^{-\sigma^2 T/8}$, $\mathbb{E}[\mathrm{LVR}_T] = V_0(1 - e^{-\sigma^2 T/8})$ exactly.
- Range $[P/w, Pw]$ centred on the price, $x = L(P^{-1/2} - P_b^{-1/2})$: same rate $\frac{\sigma^2}{4} L\sqrt{P}$, but $V = 2L\sqrt{P}(1 - w^{-1/2})$. Hence

$$\frac{\mathrm{LVR}}{V} = \frac{\sigma^2}{8\,(1 - w^{-1/2})} \quad \text{per unit time, while the price is in range.}$$

The second formula is derived here from the first; simulation agrees within 2% (test suite).

# Backup: queue position model

One unit sell order at the best ask; $a$ orders ahead, $b$ behind, $c$ at the best bid; one-tick spread.

| Event | Rate ($s^{-1}$) |
|---|---|
| market buy (front of ask queue; fills us if $a = 0$) | $\mu$ |
| limit sell joins behind us | $\lambda$ |
| cancellation ahead / behind | $\theta a$ / $\theta b$ |
| limit buy joins bid / bid depletes by one | $\lambda$ / $\mu + \theta c$ |
| informed sweep up (fills everyone, fair value $+J$) / down (price moves away) | $\nu$ / $\nu$ |

Payoff per fill, in ticks against the mid after the next price change: $\frac{1}{2} - \mathrm{AS}$ with $\mathrm{AS} = 1$ if the fill empties the queue, $2P_{\uparrow}(c, b) - 1$ otherwise, and $J$ for a sweep. Solved exactly as an absorbing Markov chain; truncation and Monte Carlo checks in the tests.

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